

The features of GroudSim are listed below:

- It is a powerful Java based simulation tool kit for scientific applications. It combines grid and cloud infrastructures, and is based on a discrete-event simulation tool kit.
- Improved performance as compared to process based approaches in other simulators.
- Can be extended easily by adopting probability distribution packages.

The most unique feature in GroudSim is GroundEntity, which has its own definition for error behaviour. The user can change this configuration during each error occurrence.

DCSim (Data Centre Simulation)

DCSim is regarded as an extensible data centre simulator designed in Java. It provides a stable and easy framework for developing and performing high-end experiments on data centre management techniques and algorithms. DCSim, being an event-driven simulator, simulates a data centre IaaS offering to multiple clients.

DCSim provides the additional capability of modelling replicated VMs, sharing incoming workloads as well as dependencies between VMs that are part of a multi-tiered application.

The features of DCSim are listed below:

- Contains a multi-tier application model that allows the

simulation of dependencies between VMs.

- Facilitates rapid development, evaluation and feedback on data centre management policies and algorithms. 

References

- <http://www.cloudbus.org/cloudsim/>
- Wickremasinghe, B., & Buyya, R. (2009). CloudAnalyst: A CloudSim-based tool for modelling and analysis of large scale cloud computing environments. MEDC project report, 22(6), 433-659.
- <https://greencloud.gforge.uni.lu/>
- <http://www.arcos.inf.uc3m.es/~icancloud/Home.html>
- <http://www.cloudbus.org/cloudsim/emusim/>
- Ostermann, S., Plankensteiner, K., Prodan, R., & Fahringer, T. (2010, August). GroudSim: An event-based simulation framework for computational grids and clouds. In European Conference on Parallel Processing (pp. 305-313). Springer Berlin Heidelberg.
- Keller, G., Tighe, M., Lutfiyya, H., & Bauer, M. (2013, May). DCSim: A data centre simulation tool. In IM (pp. 1090-1091).

By: Prof. Anand Nayyar

The author is assistant professor in the department of computer applications and IT at KCL Institute of Management and Technology, Jalandhar, Punjab. He loves to work on open source technologies, embedded systems, cloud computing, wireless sensor networks and simulators. He can be reached at anand_nayyar@yahoo.co.in.

Continued from Page 36...



Figure 11: Best practices for cloud management

There are technologies that ease the management of the cloud in very huge projects, where management becomes a really tough task. Imagine monitoring and scaling 30,000+ servers behind a shopping website on a regular basis, with this provisioning of servers increasing steeply during sales and offers provided by the sites. Some cloud providers offer dynamic provisioning, while some provide these services on-demand. Generally, there are three types of provisioning supported by cloud providers, which are shown in Figure 10.

A good management of your cloud environment not

only ensures the security of your cloud but also plays a key role in using your resources in a very wise, effective and economical way.

Some pointers which enable good management of cloud services are shown in Figure 11.

There are no plug-and-play devices available to ensure the 100 per cent security of anything in this world. The only thing we can do is to take the necessary precautions to prevent damage. We humans are the biggest threat to this world and to cloud security. So, remember, it's you who can make your security air tight. 

References

- https://en.wikipedia.org/wiki/Cloud_computing_security
- <https://aws.amazon.com/security/>
- https://en.wikipedia.org/wiki/Cloud_management

By: Amit Ranjan

The author has been working as an associate cloud architect in an MNC for over four years. He generally explores all things related to the cloud. He is proficient with Amazon AWS, and has experience on server side Java and Web development modules, including services and server deployment.